Google Cloud

Overview of cloud architeture

Hi there!

In this video we're going to dive into cloud architecture.

And to do that, we're heading to the shopping mall!

Ok, so picture a huge shopping mall with all kinds of stores, each selling different products and services.

Whatever you need, you'll likely find it.

The best part?

As a shopper, you have access to all of these cool things without having to worry about the building's utilities, maintenance, or security.

The mall handles all of that.

Well, cloud architecture is like a shopping mall.

Thinking about this in business, an organization may need to store files.

That's no problem!

Need a lot of computing power?

Easy as can be.

Want to analyze data?

Right this way.

And behind-the-scenes activities are up to the cloud provider.

The organization chooses the services they need, and the cloud provider puts the services to work.

Cloud architecture is what builds the cloud.

And how the various components and technologies are arranged is what lets organizations pool, share, and scale resources over a virtual network.

The components that make up the structure include a frontend platform, a backend platform, a cloud-based delivery model, and a network.

Frontend platforms include the parts of the architecture that users interact with: the screen design, apps, and home or work internet networks.

Frontend platforms allow users to connect and use cloud computing services.

For example, say a user opens the web browser on their mobile phone and edits a Google Doc.

In this case, the browser, phone, and Google Docs app are all frontend cloud architecture components.

In contrast, backend platforms are the components that make up the cloud itself, including computing resources, storage, security mechanisms, and management.

A primary backend component is the application.

This behind-the-scenes software is accessed by a user from the frontend and then the backend component completes the task.

When you use an online shopping app, you may browse the products first, add them to your cart, and then make a purchase.

All these actions are possible because the backend application is at work behind the scenes to complete the shopping experience.

Another key backend component is service.

You can think of service as the heart of a cloud architecture because it takes

care of all the tasks happening in the computing system and manages the resources users access.

The third primary backend component is runtime cloud.

Runtime cloud provides a space for all cloud services to perform efficiently.

It’s similar to your laptop or phone’s operating system and ensures cloud services run smoothly.

Runtime cloud technology creates monitors for all services, like applications, servers where data is stored, space for storing files, and network connections.

Next, we have storage.

This is where cloud service providers keep the data to run systems.

Different cloud service providers offer flexible and scalable storage designed to hold and organize vast amounts of data in the cloud.

Infrastructure is probably the most well-known component of cloud architecture.

It’s made up of important hardware that allows the cloud to work, and it keeps systems at their

best, including the central processing unit or CPU, the graphics processing unit or GPU, and the network devices.

The final primary backend component is security.

Because more and more organizations are adopting cloud computing, it’s essential to ensure that everything is kept safe.

Planning and designing security for data and networks provides critical supervision of systems, stops data loss, and avoids downtime.

Cloud architecture is the blueprint of how all of these components can deliver a highly agile and scalable solution.

Understanding cloud architecture is an important step toward becoming a skilled cloud professional, and you’re well on your way!